

DATA ANALYST INTERNSHIP

Task 5: Exploratory Data Analysis (EDA)

Mall Customers Dataset — EDA Report of Findings

Dataset	Mall Customers Dataset
Total Records	200 customers
Total Features	5 (CustomerID, Gender, Age, Annual Income, Spending Score)
Missing Values	None (Clean Dataset)
Tools Used	Python, Pandas, NumPy, Matplotlib, Seaborn

Objective

Perform Exploratory Data Analysis (EDA) on the Mall Customers dataset to uncover patterns, trends, correlations, and anomalies using statistical summaries and a comprehensive suite of visualizations. The goal is to understand customer demographics and spending behavior to derive actionable business insights.

1. Data Loading and Inspection

1.1 Dataset Structure

The dataset was loaded using Pandas. It contains 200 rows and 5 columns with no missing values, making it a clean dataset ready for analysis.

Column	Data Type	Description
CustomerID	int64	Unique identifier for each customer
Gender	object	Customer gender (Male / Female)
Age	int64	Customer age in years
Annual Income (k\$)	int64	Annual income in thousands of dollars
Spending Score (1-100)	int64	Score assigned by the mall (1 = low, 100 = high)

1.2 Statistical Summary (.describe())

Statistic	Age	Annual Income (k\$)	Spending Score (1-100)
Count	200	200	200
Mean	38.85	60.56	50.20
Std Dev	13.97	26.26	25.82
Min	18	15	1
25th Pct	28.75	41.50	34.75
Median	36.0	61.50	50.0
75th Pct	49.0	78.0	73.0
Max	70	137	99

1.3 Gender Distribution

Female: 112 customers (56%) | Male: 88 customers (44%) — The dataset is slightly female-dominant.

2. Univariate Analysis

2.1 Histograms — Distribution of Numerical Features

Histograms reveal the frequency distribution of each numerical variable, helping identify skewness, spread, and data concentration zones.

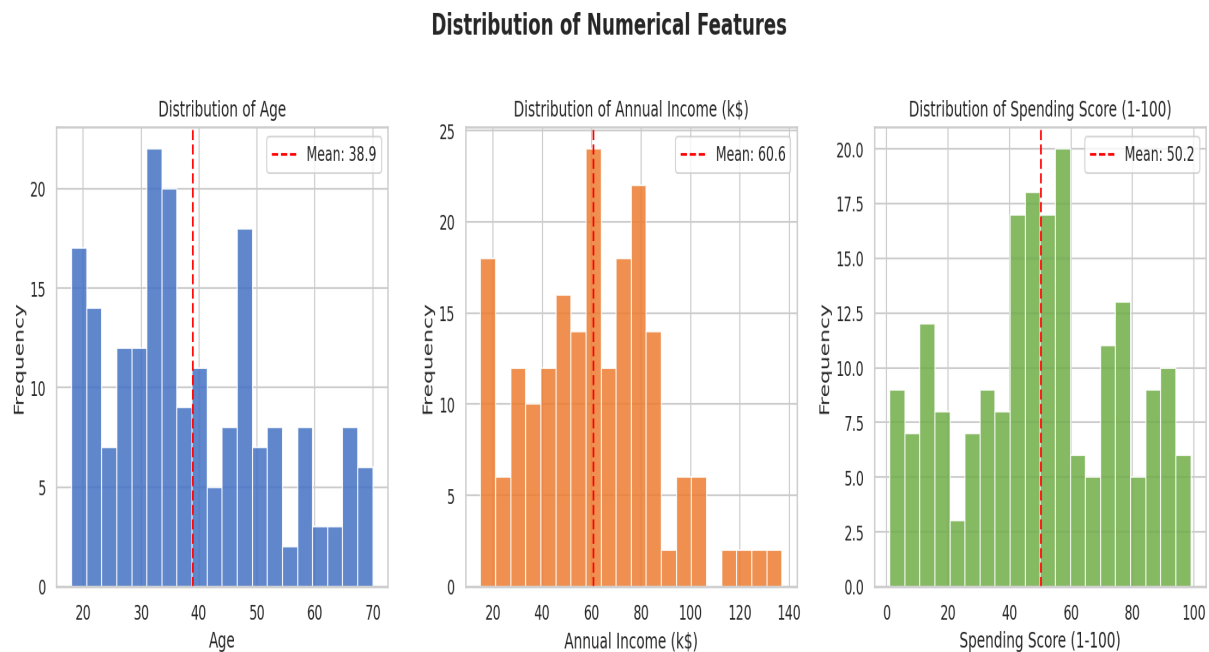


Figure 1: Histograms of Age, Annual Income, and Spending Score

■ Observations:

- **Age:** Slightly right-skewed. Most customers fall between 20–40 years, with a peak around 30. Older customers (60+) are fewer.
- **Annual Income:** Approximately normal distribution centered around \$60k. A few high-income outliers exist above \$120k.
- **Spending Score:** Roughly uniform with slight peaks at extreme ends (very low and very high spenders), suggesting polarized spending behavior.

2.2 Gender Distribution — Pie Chart

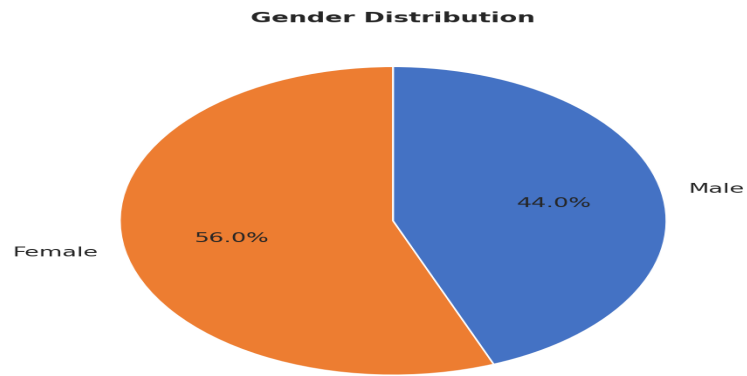


Figure 2: Gender Distribution of Mall Customers

■ Observations:

- The dataset has 56% Female and 44% Male customers. While not heavily imbalanced, gender-based segmentation is still meaningful for behavioral analysis.

3. Bivariate Analysis

3.1 Boxplots — Outlier Detection by Gender

Boxplots display the distribution of each feature split by gender, showing median, IQR, and any outliers.

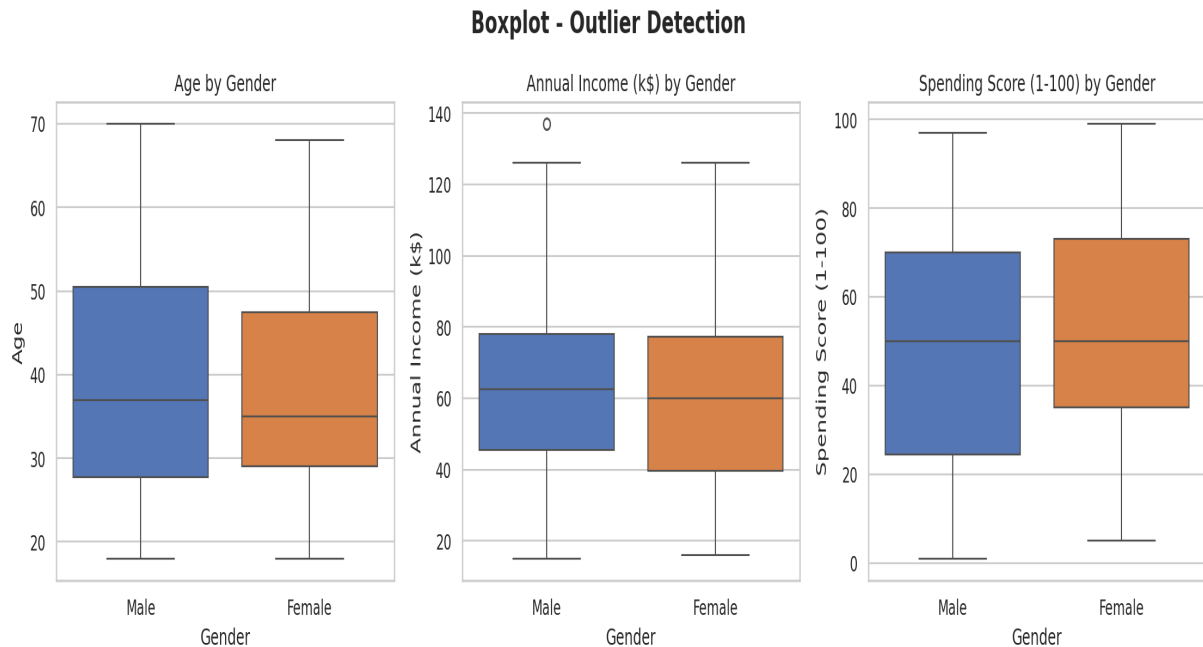


Figure 3: Boxplots comparing Age, Income, and Spending Score by Gender

■ Observations:

- **Age:** Males and females have similar age distributions. A few outliers appear in both groups above age 60.
- **Annual Income:** Males show a slightly higher median income (~\$62k) vs females (~\$59k). Notable high-income outliers exist in both groups.
- **Spending Score:** Females show a marginally higher median spending score than males, suggesting females are slightly more active spenders in this mall.

3.2 Scatter Plots — Relationship Between Variables

Scatter plots help visualize linear or non-linear relationships between pairs of numerical features, colored by gender to reveal group-level patterns.

Scatter Plots - Relationship Analysis



Figure 4: Scatter Plots — Age vs Spending Score & Income vs Spending Score

■ Observations:

- **Age vs Spending Score:** Clear negative trend — younger customers (18–35) tend to have higher spending scores. Customers aged 40+ cluster around moderate-to-low spending scores. This suggests youth-driven consumption patterns.
- **Income vs Spending Score:** No strong linear trend overall, but visible clusters emerge: (1) Low income + low spending, (2) Low income + high spending, (3) High income + low spending, (4) High income + high spending — classic customer segmentation.
- Gender does not strongly differentiate spending behavior in scatter analysis; both genders appear in all clusters.

4. Multivariate Analysis

4.1 Correlation Heatmap

The heatmap displays Pearson correlation coefficients between all numerical features, ranging from -1 (perfect negative) to +1 (perfect positive).

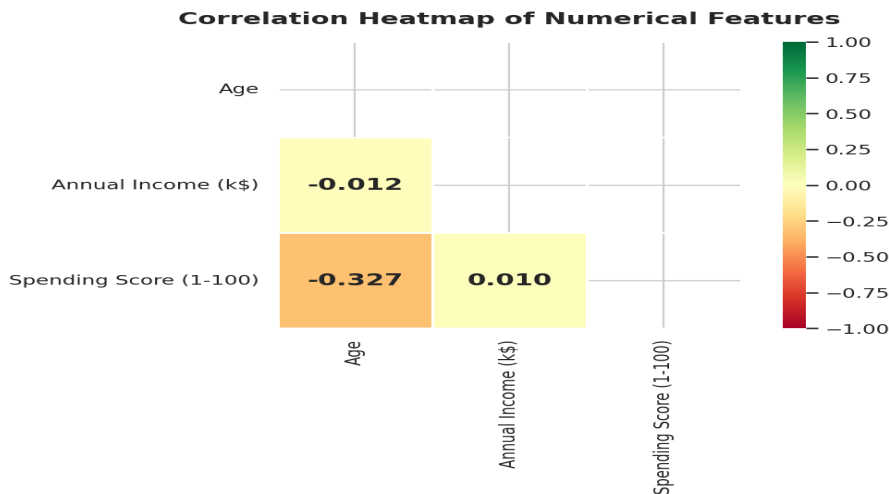


Figure 5: Correlation Heatmap of Numerical Features

■ Observations:

- **Age vs Spending Score: -0.327** — Moderate negative correlation. As age increases, spending score decreases. This is the strongest relationship in the dataset.
- **Age vs Annual Income: -0.012** — Virtually no correlation. A customer's age does not predict their income level.
- **Income vs Spending Score: +0.010** — Near-zero correlation. Higher income does not directly translate to higher spending in this mall.
- No multicollinearity detected among the features — they are largely independent, which is ideal for downstream modeling.

4.2 Pairplot — Pairwise Relationships

The pairplot provides an all-in-one view of pairwise relationships (scatter plots) and individual distributions (KDE on diagonal), segmented by gender.

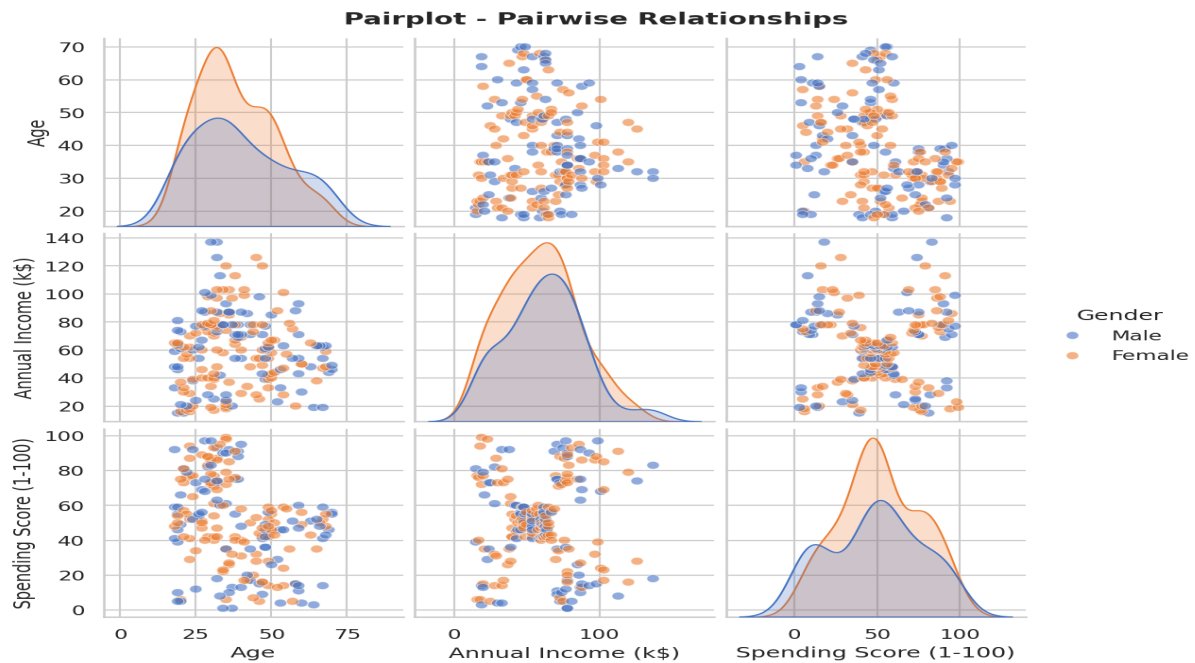


Figure 6: Pairplot — All Pairwise Feature Relationships by Gender

■ Observations:

- KDE plots on the diagonal confirm the distributions seen in histograms — Age is slightly right-skewed, Income is close to normal, Spending Score is near-uniform.
- The Age vs Spending Score scatter confirms the negative relationship consistently across both male and female segments.
- The Income vs Spending Score panel reveals distinct cluster formations — potential for k-means customer segmentation.
- Female customers tend to cluster slightly higher on spending scores compared to males, visible in the KDE overlays.

5. Summary of Findings

Customer Demographics	200 customers aged 18–70, with 56% female. Most customers are between 20–40 years old. The typical customer earns ~\$60k/year with a spending score around 50.
Age & Spending Behavior	Age negatively correlates with spending score ($r = -0.327$). Younger customers are the most active spenders. Mall campaigns targeting the 20–35 age group are likely most effective.
Income vs. Spending	Annual income has almost no correlation with spending score ($r = 0.010$). This is a key insight — income alone is not a reliable predictor of spending behavior.
Customer Clusters Identified	Scatter plots reveal at least 5 natural customer segments: Low/Medium/High income groups with varying spending profiles — ideal candidates for K-Means clustering (Task 6).
Outliers	A few high-income customers (>\$120k) exist. Some customers over 60 continue active spending, though they are a minority.
Gender Insights	Females slightly outperform males in spending scores. No extreme gender-based income gap was observed.
Data Quality	The dataset is clean — 0 missing values, appropriate data types, and reasonable ranges for all features.

6. Interview Questions & Answers

Q1. Which plots do you use to check correlation?

Correlation Heatmap (shows exact Pearson coefficients between all pairs) and Pairplot (shows scatter plots for all feature combinations). Scatter plots for specific pairs are also used to visually confirm the direction and strength of relationships.

Q2. How do you handle skewed data?

Apply log transformation (`np.log1p`) for right-skewed data, or square-root transformation for moderate skew. For left-skewed data, square or cube transformations help. After transformation, re-check distributions using histograms and Q-Q plots to confirm normalization.

Q3. How to detect multicollinearity?

Use the correlation heatmap — values above 0.8 or below -0.8 between independent variables signal multicollinearity. For regression models, calculate the Variance Inflation Factor (VIF); $VIF > 5$ –10 indicates problematic multicollinearity.

Q4. What are univariate, bivariate, and multivariate analyses?

Univariate: Analysis of a single variable in isolation (e.g., histogram of Age). Bivariate: Relationship between two variables (e.g., scatter plot of Age vs Spending Score). Multivariate: Simultaneous analysis of three or more variables (e.g., pairplot with gender hue, 3D scatter plots).

Q5. Difference between heatmap and pairplot?

Heatmap: Displays a matrix of correlation coefficients as colored tiles — best for quickly spotting strong/weak correlations numerically. Pairplot: Shows actual scatter plots for every feature pair plus distribution plots on the diagonal — best for visualizing the shape and direction of relationships.

Q6. How do you summarize your insights?

Structure insights around: (1) Data quality check, (2) Individual feature distributions, (3) Relationships between features, (4) Outliers and anomalies, (5) Business-relevant conclusions. Always tie statistical findings back to actionable recommendations.

Q7. What is EDA and why is it important?

EDA (Exploratory Data Analysis) is the process of systematically investigating a dataset using statistics and visualizations before applying models. It is important because it: reveals data quality issues, highlights key patterns and relationships, informs feature selection, prevents modeling on flawed assumptions, and generates hypotheses for further analysis.

7. Conclusion

This Exploratory Data Analysis of the Mall Customers dataset successfully achieved its objective of extracting meaningful patterns and insights from structured customer data. The analysis confirms that **age is the strongest predictor of spending behavior** (negative correlation of -0.327), while annual income alone does not determine spending patterns. The data reveals clear customer segmentation opportunities — particularly 5 distinct clusters visible in the Income vs Spending Score scatter plot — which would be ideal for a follow-up K-Means Clustering analysis.

EDA is an indispensable first step in any data science workflow. It ensures analysts understand the data's structure, quality, and characteristics before committing to modeling approaches. The techniques applied here — histograms, boxplots, scatter plots, heatmaps, and pairplots — collectively provide a 360-degree view of the dataset and lay a strong foundation for predictive modeling or customer segmentation tasks.